

NIGEL LI (李永沛)

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EDUCATION

Columbia University

New York, NY

MA in Mathematics of Finance

09/2025 – 12/2026

- GPA: 4.07/4.0; Coursework: Numerical Methods in Finance; Stochastic Methods in Finance; Programming for Quantitative & Computational Finance; Financial Risk Management & Regulation; Machine Learning for Finance; Multi-Asset Portfolio Management; Hedge Fund Strategies
- Activities: Selected Participant, UBS Shanghai FinAI Tomorrow Talent Program; Roles: Graduate Teaching Assistant (Aug 2026 – Present).

University of New South Wales

Sydney, Australia

Dual Degree B. Commerce (Business Analytics) & B. Computer Science (Artificial Intelligence)

02/2021 – 08/2025

- GPA: 81.8/100; High Distinction (HD) in Business School | Distinction (D) in School of Engineering
- Coursework: Introductory Econometrics (Rank: 2/223); Quantitative Business Analytics (Rank: 1/125); Financial Management; Game Theory & Business Strategy; Discrete Mathematics; Machine Learning & Data Mining; Neural Networks & Deep Learning; Algorithm Design & Analysis; Data Structures & Algorithms
- Honors: Engineering Dean's Honours List (2021–23) | Business School Dean's Award | Research Scholarship (2022)
- Activities: Finalist, FMAA x SIG (Susquehanna) Financial Showcase (2023); Finalist, CommBank Global Markets Case Competition (2023).
- Programs: Akuna Capital Options 101. Roles: Undergraduate Teaching Assistant (Feb 2023 – May 2026).

PUBLICATIONS

- N. Li, Y. Han. "Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL." *7th ACM International Conference on AI in Finance (ICAIF '26)*, 2026. Accepted.
- N. Li (Captain), A. R. Kommareddy, A. V. Kramarenko, W. Y. Qiu, L. K. Y. Tam, J. Wang. "When '\$1' Isn't One: Stablecoin Discounts and Market Fragmentation." *IAQF Academic Affiliate Membership Student Competition*, 2026. [Winning paper](#).

WORK EXPERIENCE

HSBC

Hong Kong, SAR

Global Markets Summer Analyst / Wealth Solutions & Structured Credit Trading (Return Offer)

06/2026 – 08/2026

- Developed an Excel-based credit-derivatives pricing project emulating core Bloomberg SWPM and CDSW functionality, including KRW swap-curve bootstrapping, interest-rate swap valuation, tenor-based DV01 hedge analytics, CDS pricing, and First-to-Default basket modelling.
- Engineered a dynamic BQL-enabled Excel comparables framework to evaluate Credit-Linked Notes against reference cash and callable bonds, incorporating instrument-specific eligibility constraints and relative-value metrics.
- Automated a daily risk report (CS01, DV01, logvega) across the desk's live Callable CLN book via VBA, aggregating live positions and risk into a single refreshable report in place of manual re-pricing.

Challenger Investment Management

Sydney, Australia

Summer Investment Analyst Intern / ALM & Liquids Trading, Credit (Return Offer)

11/2024 – 02/2025

- Developed and proposed a delta-neutral crypto basis strategy yielding ~9% annualized returns at a 1.4 Sharpe ratio, as measured by a comprehensive execution and risk management framework adopted by the desk, through quantitative modeling of term-structure dynamics, slippage, and historical drawdown constraints.
- Delivered credit assessments on private and unrated fixed-income issuers across IG and HY, as measured by integration into investment memos and internal risk scoring adopted by portfolio managers, by applying Moody's methodology.
- Strengthened ALM monitoring across mortgage trailing-interest and life-insurance portfolios, as measured by stress reports adopted into desk reporting cycles, by building cashflow projection models and running base-case versus worst-case rate and credit-spread shock scenarios calibrated to historical drawdowns.

Wellington Management

Sydney, Australia

InvesTech Intern (Return Offer)

03/2024 – 10/2024

- Built a systematic FX pairs screening framework across G10 and EM crosses, as measured by adoption into a PM's pair-selection workflow for relative-value trades, by implementing a cointegration matrix and the Cointegrated Augmented Dickey-Fuller (CADF) test across FX and commodities.
- Delivered a *firm-wide recognized* investment memo automation tool with sentiment analysis, as measured by ~20% reduction in analyst drafting time and presentation to Directors/Board, by architecting a Retrieval-Augmented Generation (RAG) pipeline over the firm's internal research archive.

SKILLS & ADDITIONAL INFORMATION

Computer: Bloomberg Terminal (BQL), Excel/VBA, Python, SQL, R, Java, FactSet, Capital IQ, Tableau, Microsoft Office

Extracurricular Activities: UNSW Personal Investment Society (Vice President Externals), Taiwanese Student's Association (Director)

Languages: English (Native), Mandarin (Native), Cantonese (Fluent)

Interests: Skiing, Basketball, Chess, Hiking, Travelling